

MUHAMMAD ALI ABID

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PROFESSIONAL SUMMARY

AI Engineer and Backend Developer with hands-on experience building production-grade data systems, LLM-powered applications, and scalable backend pipelines for government agencies and international clients. Developed AI applications leveraging OpenAI API and LangChain, engineered ETL pipelines and PostgreSQL-backed systems handling national-scale datasets, and built REST APIs with FastAPI for data-driven products. Delivered the Trade Intelligence Portal for Pakistan Bureau of Statistics and healthcare analytics solutions for US-based clients. Core expertise spans LLM integration, backend development, geospatial analytics, and end-to-end ML pipelines with a strong foundation in Python, SQL, and cloud-native deployment.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Statsmodels, Scipy, Power BI, DAX

ML/AI/NLP: Scikit-learn, BERT, Hugging Face Transformers, OpenAI API, LangChain, Time Series (ARMA/ARIMA), LDA Topic Modeling

Databases: PostgreSQL, SQL Server, MySQL, PostGIS, QGIS, Leaflet, Geospatial Analytics, MongoDB, Redis, AWS RDS

Tools & Platforms: n8n, Gradio, Jupyter, Git

Frontend: React, NextJs, Streamlit

Backend: FastAPI, NextJs, Flask

Cloud & DevOps: Vercel, Netlify, DigitalOcean, Docker, AWS EC2, ECS

PROFESSIONAL EXPERIENCE

Data Scientist / Data Engineering Intern

Jan 2026 – Feb 2026

Pakistan Bureau of Statistics (PBS) | Islamabad, Pakistan

- Engineered Government Trade Intelligence Portal handling national import/export data for statistical analysis and policy planning
- Integrated PostgreSQL database with pgAdmin4; designed optimized schemas for large-scale trade datasets
- Implemented GIS solutions using QGIS and Leaflet for geospatial visualization of trade routes, port analytics, and regional patterns
- Developed Power BI dashboards with hierarchical drill-downs presenting trade metrics, commodity flows, and trend analysis
- Collaborated with statistical officers to transform raw customs data into actionable intelligence for government reporting

Associate Data Analyst

May 2025 - August 2025

Ambient Cloud | Islamabad, Pakistan

- Developed AI-enabled, cloud-based product features supporting data-driven decision making for enterprise clients
- Designed and implemented automation workflows using n8n for streamlined data pipelines and task orchestration
- Built and maintained ETL pipelines for data ingestion, preprocessing, and transformation across multiple data sources
- Created analytical dashboards and reports for internal monitoring and business intelligence
- Collaborated with cross-functional teams integrating automation, analytics, and cloud workflows

Trade Intelligence Portal

Government of Pakistan - Enabling data-driven trade policy decisions for national economic planning | PostgreSQL, QGIS, Leaflet, Power BI, TypeScript

- Built end-to-end trade data visualization platform for Pakistan Bureau of Statistics managing national customs data
- Integrated geospatial layer enabling policymakers to analyze regional trade patterns, port throughput, and commodity flows
- Developed hierarchical analytics dashboards supporting evidence-based decisions for trade agreements and tariff policies

ModivCare Trip Tracker Pro

USA Healthcare Sector - Improving medical transportation access for patients requiring non-emergency medical trips | TypeScript, REST APIs, Analytics Dashboard

- Engineered healthcare transportation tracking system for US-based ModivCare serving patients with mobility challenges
- Implemented real-time trip monitoring reducing missed appointments and improving healthcare accessibility
- Delivered operational efficiency dashboards helping dispatch teams optimize routes and reduce patient wait times

Stock Return Forecasting System

Retail Investors & Financial Analysts - Democratizing quantitative finance tools for investment decisions | Python, ARMA, Statsmodels, Vercel

- Built ARMA(1,1) model forecasting monthly stock returns capturing autocorrelation and market shock persistence
- Deployed live web application on Vercel enabling retail investors to access institutional-grade forecasting tools
- Applied econometric techniques making quantitative analysis accessible to non-technical investors

Layer Forge AI

Content Creators & Developers - Streamlining creative workflows through AI-powered generation | TypeScript, OpenAI API, LangChain

- Developed AI application leveraging LLM capabilities for intelligent content generation and assistance
- Integrated OpenAI API with custom prompt engineering enabling non-technical users to harness AI capabilities

Fine-Tuning BERT for Sentiment Analysis

NLP Research Community - Advancing understanding of transformer-based text classification methods | Python, BERT, Hugging Face Transformers, PyTorch

- Conducted comprehensive 55+ page research on transformer architecture and contextual embeddings
- Implemented end-to-end NLP pipeline: preprocessing, tokenization, fine-tuning, and model evaluation
- Contributed detailed analysis of BERT performance characteristics for sentiment classification tasks

Multi-Platform Media Content Analysis

CALD Communities in Australia - Uncovering media representation patterns affecting immigrant communities | Python, LDA Topic Modeling, VADER Sentiment Analysis, NLP

- Analyzed 1,500+ media records across Factiva, Twitter/X, and Reddit examining representation of diverse communities
- Implemented LDA topic modeling revealing dominant narratives and potential bias in mainstream media coverage
- Provided statistical evidence of platform-specific framing patterns informing media literacy and advocacy efforts

AI Minbar - Islamic Super App

Global and Local Muslim Community - Providing unified access to Islamic resources for religious education and practice | TypeScript, Next.js, Full-Stack Development

- Built comprehensive application featuring Qur'an, Hadith search, prayer times, and masjid location services
- Implemented search and bookmarking helping users navigate religious texts for study and daily practice
- Designed accessible interface serving Muslims worldwide regardless of technical expertise or language background

AI Burhan Portal

Islamic Educational Community - Providing centralized access to Islamic scholarship and verified religious content | CSS, HTML, JavaScript, Web Development

- Developed Islamic educational portal aggregating scholarly resources, articles, and verified religious content
- Built content management system enabling scholars to publish and organize educational materials
- Designed user-friendly interface making authentic Islamic knowledge accessible to global audience

Elite Performance Garage - Digital Transformation

Automotive Business (Client) - Complete digital onboarding transitioning physical garage operations to online presence | HTML, CSS, JavaScript, Full-Stack Web Development

- Led complete digital transformation for automotive garage, transitioning from physical-only to online-enabled business
- Developed professional website with service catalog, appointment booking, and customer inquiry system
- Implemented SEO-optimized content strategy increasing online visibility and customer acquisition for local business

Retail Management System (RMS)

Small & Medium Retail Businesses - Digitizing operations for shops lacking enterprise software budgets | HTML, CSS, JavaScript, Full-Stack

- Built full-stack retail system handling inventory, sales tracking, and basic reporting for small businesses
- Provided affordable digital solution helping local retailers compete with chain stores through better operations

EDUCATION

Bachelor of Science in Business Data Analytics (FinTech Major)

COMSATS University Islamabad | 2023 - 2027 | GPA: 3.19/4.00

CERTIFICATIONS

- IBM Certified Data Engineer
- Meta Certified Frontend Developer
- Google Certified Business Analyst
- Data Analytics & Business Intelligence - Digital Skills Training Program (DSTP)
- Freelancing Professional Development - Digital Skills Training Program (DSTP)